

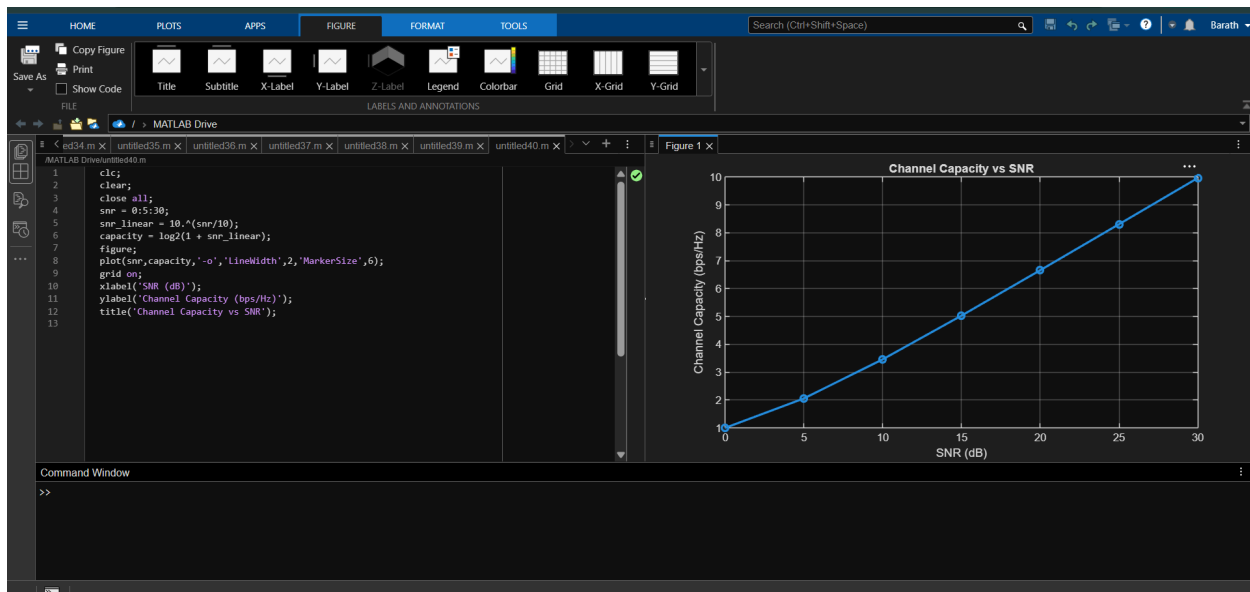
Experiment: 9 Massive MIMO Capacity vs SNR and Gain Using MATLAB

Objective – 1

MATLAB Code:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

OUTPUT:

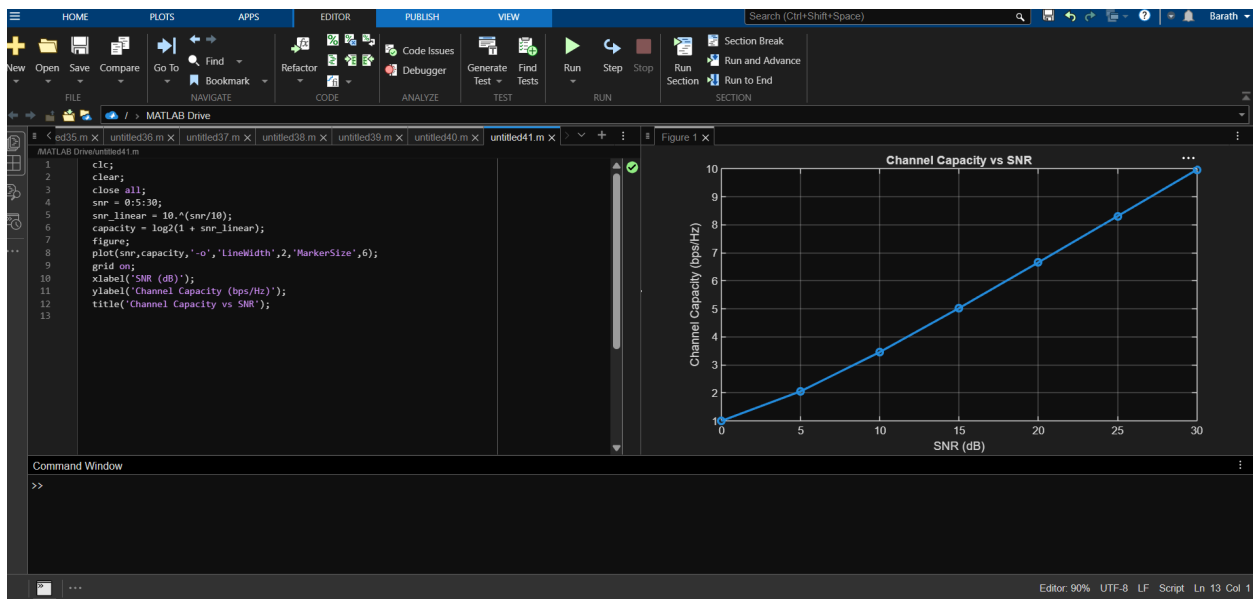


Objective – 2

MATLAB Code:

```
clc;
clear;
close all;
snr = 0:5:30;
snr_linear = 10.^(snr/10);
capacity = log2(1 + snr_linear);
figure;
plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);
grid on;
xlabel('SNR (dB)');
ylabel('Channel Capacity (bps/Hz)');
title('Channel Capacity vs SNR');
```

OUTPUT:



Objective – 3

MATLAB Code:

```
clc;
clear;
close all;
antennas = [4 8 16 32 64];
```

```

gain = 10*log10(antennas);
figure
bar(antennas, gain)
grid on
% Number of Antenna Elements
% Antenna Gain (dB)
xlabel('Number of Antenna Elements')
ylabel('Antenna Gain (dB)')
title('Antenna Gain for Different Antenna Array Sizes')

```

OUTPUT:

